ROBERT GORDON UNIVERSITY ABERDEEN

School of Computing

MSc Project Report

Building a recommendation model for ecommerce platform using customer online interaction data and purchase history.

Joseph Debrah

This report is submitted as part of the requirements for the degree of

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at Robert Gordon University, Aberdeen, Scotland

I confirm that the work contained in this Honours project report has been composed solely by myself and has not been accepted in any previous application for a degree. All sources of information have been specifically acknowledged and all verbatim extracts are distinguished by quotation marks.

Student Signature: Joseph Debrah **Date: 22/08/2024**

Abstract

The covid-19 pandemic had a huge impact on various aspect of global economy, one of the most notable changes is the exponential increase in ecommerce business. It significantly influenced the way people shopped, with many preferring the safety and convenience of online shopping. Even those who were initially hesitant began to do so out of necessity, leading to a wider acceptance and trust in ecommerce.

Recommendation systems have played a key role in the success of ecommerce by providing personalised products and services enhancing customer experience and increasing sales. The purpose of this research is to build a recommendation model with the goal of enhancing user experience and boosting sales in ecommerce business. Our model utilised datasets consisting of customer online interaction and their purchase history. The model employs the memory based collaborative filtering system and KNN algorithm.

Collaborative filtering system is one of the most extensively used technologies for recommendation systems, as it is an improved intelligent searching mechanism for recommending personalised items. Additionally, our model employed the Euclidean distance method to find similarities between users. It also touches on the process involved in achieving the optimal value of K, which is the number of nearest neighbours the algorithm should consider in making recommendations.

Furthermore, the paper discusses the various types of recommendation systems and the various techniques used. It also delves into research trends in recommendation as well as real life application of recommendation systems using Amazon and Netflix as case study. For each website we give a summary of the recommendations provided, the type of technology used and the information gathered from users or customers to make recommendations.

Lastly, a recommendation engine was developed that takes a user email as input, checks if the users exit in the dataset, identify similar users based on their online interaction data. It retrieves items purchased by the similar users and sort out items that has not already been purchased by the target user. It then returns top items as final recommendations based on their frequency among the similar users. The model was also evaluated using Mean Average Precision (Map) to measure how well the model predict relevant items and how accurately it can rank them.

Acknowledgements

I would like to dedicate this research to the Almighty God, whose blessings and guidance have been the foundation of my success.

To my beloved family and mom Alice, whose unconditional love and unwavering support have been the driving force behind all my achievements. Your prayers and encouragement have been the wind beneath my wings, propelling me forward every step of the way.

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Lastly, to my cherished A.K your presence in my life has been a true blessing and I am grateful for being there for me through thick and thin.

This research is a testament to the guidance, support, and love that each of you has so generously provided

Declaration

I confirm that the work contained in this MSc project report has been composed solely by myself and has not been accepted in any previous application for a degree. All sources of information have been specifically acknowledged and all verbatim extracts are distinguished by quotation marks.

Signed Joseph Debrah. Date 22nd August 2024.

Table of Contents

Abstract i
Acknowledgements iii
Declaration iv
Chapter1 1
1.1 Introduction 1
1.2 Background 1
1.3 Problem Statement 2
1.4 Project Specification 3
1.4.1 Project Aim 3
1.4.2 Project Objectives 3
1.5 Project Report 4
1.5.1 Structure 4
Chapter 2 5
2.1 Literature Review 5
2.2 Overview of Recommendation Systems 5
2.3 Phases of Recommendation 5
2.4 Recommendation Systems (RS) in Ecommerce 6
2.5 Types of Recommendation Systems 6
2.5.1 Collaborative Filtering Recommendation System 7
2.5.2 Advantages of Collaborative Filtering 8
2.5.3 Disadvantages of Collaborative Filtering 9
2.5.4 Content-Based Recommendation System 9
2.5.5 Demographic System 9
2.5.6 Knowledge Based System. 10
2.5.7 Hybrid System 10
2.5.7.1 Combination Strategies of the Hybrid System 11
2.5.7.2 Weighted 11
2.5.7.3 Mixed 11
2.5.7.4 Switching 11
2.5.7.5 Feature Combination 12
2.6 Recommendation Techniques 12
2.6.1 Types of recommendation Techniques 13
2.6.2 Text Mining 13
2.6.3 K-NN (K-Nearest Neighbour) 14

2.6.4 Clustering 15
2.6.5 Matrix Factorisation 15
2.7 Research Trends of Recommendation System Techniques 16
2.8 Real Applications of Recommendation Systems. 18
2.8.1 Amazon 18
2.8.2 Netflix 20
2.9 Evaluation Metrics 21
2.9.1 Recall and Precision 22
2.9.2 Accuracy 23
2.9.3 ROC Curve 24
2.9.4 Recall 25
2.9.5 Mean Average Precision (MAP) 25
2.10 Future Directions in Recommendation System 26
Chapter 3 27
3.1 Methodology 27
3.2 Data Understanding 27
3.3 Data Preparation 29
3.3.1 Cleaning and normalising the numerical features 29
3.4 Model Selection 30
3.5 Legal, Ethical and Professional issues 31
3.5.1 Legal Considerations - Data Usage and Privacy. 31
3.5.2 Ethical and Professional issues 32
3.6 Chapter Summary 33
Chapter 4 34
4.1 Implementation 34
4.2 Visualisation and Statistical data analysis in Python 34
4.4 Determining the Best Value of K 40
4.5 Recommendation Function using collaborative filtering and k=5 41
4.6 Conclusions 43
Chapter 5 44
5.1 Results and Discussion. 44
5.2 Recommendation Engine 44
5.3 Mean Average Precision (MAP) 45
5.4 Conclusion 46
Chapter 6 47
6.1 Conclusions and Future Works 47

6.2 Conclusion 47
6.2.1 Model Optimisation and Cross-validation 47
6.2.2 Feature Selection 47
6.2.3 Personalised Recommendations 47
6.2.4 Cold Start Handling 48
6.3 Future 48
Appendices 53

List of Tables

Figure 1 Recommendation system techniques 7
Figure 2 Collaborative filtering system (Sinha and Dhanalakshmi 2019) 8
Figure 3 Hybrid System (Fayyaz et al. 2020) 10
Figure 4 recommendation techniques 13
Figure 5 Text mining process 14
Figure 6 15
Figure 7 Trends in Recommendation (Ko et al. 2022) 16
Figure 8 ratings page of amazon 19
Figure 9 Example of Netflix top 10 rows ((Molina and Bhulai 2018) 20
Figure 10 Similarity used to provide recommendations (Molina and Bhulai 2018) 21
Figure 11 22
Figure 12 A confusion matrix for a recommendation system (Fayyaz et al. 2020) 23
Figure 13 24
Figure 14 24
Figure 15 25
Figure 16 customer data 27
Figure 17 purchase data 28
Figure 18 30
Figure 19 normalising all numerical features 30
Figure 20 Time on App vs Yearly Amount Spent 35
Figure 21 Time on Website vs Yearly Amount Spent 36
Figure 22 Length of Membership vs Yearly Amount Spent 37
Figure 23 Correlation heatmap 38
Figure 24 High spenders 39
Figure 25 Low spenders 39
Figure 26 unified dataset 40
Figure 27 k value using cross validation 41
Figure 28 recommendation engine 44
Figure 29 Mean Average Precision 45
Figure 30 45

Chapter1

1.1 Introduction

This chapter covers the project background, aims & objectives. It also describes the structure of the rest of report.

1.2 Background

E-commerce is defined as the technique and methods needed to facilitate the integration of electronic communication into the business climate(Hussien, Rahma and Wahab 2021) Ecommerce systems have experienced a notable growth in sales volume in recent years, particularly due to the rapid advancement of technology and expansion of online shopping site like Alibaba and Amazon (Lee, H. I. et al. 2020). One way to enhance the shopping experience of customers is to offer product suggestions list based on the individual purchasing patterns known as recommendation system (Lee, D. and Hosanagar 2019). In an era where there is a lot of information available recommendation systems play a key role in guiding users to things they are much interested in. Recommendation systems are applicable in other fields like eBay, Flipkart, social media platforms like Facebook, online music and streaming platforms and e-learning platforms(Campos, Díez and Cantador 2014),(Linden, Smith and York 2003).

Recommendation systems have significantly helped e-commerce sites enhance their profitability. According to the data released by amazon the recommendation algorithm they implemented accounted for approximately 30% improvement in revenue. Additionally, YouTube released statistics that the engagement of users has significantly improved after implementing a robust recommendation system(Liu and Wu 2019).

Recommendation system leverages on users' behaviour and their preferences and identify similarities based on this. It then makes suggestions based on the user's interest. One of the most popular recommendation techniques is Collaborative filtering, because of its success many other algorithms have been developed based on it and they are also successful in turning out positive results.

The ultimate goal is to design a robust recommendation model that leverages on customer online interaction data such as Time Spent on App, Time Spent on Website, Yearly Amount Spent and Length of Membership against customers purchase history using K-NN algorithm. The model is then converted into a recommendation engine that can suggest relevant items to users.

1.3 Problem Statement

The exponential growth in ecommerce platforms has changed the way consumers shop by offering them access to a huge number of goods and services. However, because of the high number of options available consumers become overwhelmed and most often not decisive. To help mitigate this difficult situation recommendation system has emerge as a key tool to help enhance user experience in ecommerce by offering products and services that meet each user needs.

Also, because human needs are insatiable and the emergency of new technologies that impacts the way user interacts online, ecommerce data are now complex. This calls for more robust, scalable and efficient recommendation system. The problem this research address is to develop a recommendation system that can handle the dynamic nature of customer online interaction and overcome the limitations of the traditional systems. The ultimate goal is to offer personalised recommendation that enhances user satisfaction and improve business performance on ecommerce platforms

1.4 Project Specification

1.4.1 Project Aim

The aim of this research is to develop a recommendation model for ecommerce platform using customer online interaction data.

1.4.2 Project Objectives

The following research objectives are identified to meet the stated research aims:

Data Collection and Preprocessing:

Preprocess the data by removing irrelevant columns, handling duplicate records and standardising numerical features to ensure consistency.

Exploratory Data Analysis:

Analyse the relationship between customer interaction features (Time on App, Time on website, Length of membership and Yearly amount spent using statistical technique like correlation and covariance.

Visualise this relationship using scatter plots, correlation matrix heatmap to get insights and utilise summary statistics to identify high and low spenders.

Model Development and Evaluation:

Design a recommendation model using memory based collaborative filtering system and K-Nearest Neighbor algorithm.

Recommendation System implementation:

To Implement a recommendation engine and test it with customer data to offer items as recommendation.

1.5 Project Report

1.5.1 Structure

The remaining part of this project is presented in chapters as shown below.

Chapter Two - Literature Review: This chapter contains a detailed literature review of recommendation system models and techniques. It also explores different research that has been carried out based on peer-reviewed articles and academic journals.

Chapter Three - Methodology: This chapter covers the project scope and the methodology utilised in achieving the research goals and objectives. This includes the data preprocessing, normalising the relevant feature and model selection.

Chapter Four - Implementation: This chapter summarises how the recommendation model was designed, visualising the relevant features, selecting the optimal number of k value and implementing the recommendation engine using k-nn algorithm.

Chapter Five - Evaluation: This chapter covers discussion of results obtain from implementing the recommendation model.

Chapter Six – Conclusion: The conclusions and future works of this research is presented**.**

Chapter 2

2.1 Literature Review

This chapter analyses relevant literature surrounding ecommerce recommendation systems used in ecommerce.

2.2 Overview of Recommendation Systems

(G. Adomavicius and A. Tuzhilin 2005) defined recommendation system as a technique for filtering information that anticipates users' preferences and suggest potentially interesting products to people. Originally, recommendation systems were employed as a collaborative filtering technique to sort e-mail documents according to user relevance. The filtering process comprises of common properties like messages, replies and annotations. The term "tapestry" that supported collaborative filtering refers to an experimental system that was developed to undertake several tasks such as receiving, filtering and browsing electronic documents(Sinha and Dhanalakshmi 2019). A recommendation system job is to convert data on users' preferences to forecast users' potential future likes and interest.

Web entrepreneurs were the pioneers of the information revolution and first identified the enormous potential of recommendation system and its study is a combination of science and socio-economic life. Despite initially being an area dominated by computer scientist, today it attracts interest from psychologist, mathematics and physicists(Lü et al. 2012).

2.3 Phases of Recommendation

Recommendation systems have three main phases which are explained below:

Information collection phase: This is the first phase of the recommendation process where all relevant and key data about the users' practices and resources visited are gathered in order to create a user profile for the recommendation.

Learning Phase: In this phase an algorithm is applied which sort out the user's attribute from the data and information retrieved during the initial information gathering phase.

Prediction/ recommendation phase: This is the final phase which proposes the kind of products and services that is of interest to the customer.

2.4 Recommendation Systems (RS) in Ecommerce

The goal of RSs is to give website visitors personalized product recommendations. They gather client information and make pertinent product recommendations for the user. Recommendation systems enhance ecommerce sales in three ways:

Cross-sell: Through the recommendation of other products for the customer to buy, recommendation systems enhance cross-selling. An increase in the average order size is expected if the recommendations are sound. For example, based on the things that are already in the shopping basket, a website may suggest more products throughout the checkout process (Gorakala and Usuelli 2015).

Loyalty: Developing a loyal customer base is a crucial business strategy in the world when competitors can be reached with just a few clicks (Carlson and Sinnappan 2003). By fostering a connection of value addition between the website and the user, a recommendation system increases customer loyalty (Gorakala and Usuelli 2015).

Browsers into Buyers: A websites visitor frequently browses the content without ever making a purchase. Recommendation system can assist users in locating the goods they want to buy (Kumar et al. 2024).

2.5 Types of Recommendation Systems

Numerous research investigations have been carried out to create these recommendations systems and numerous useful systems have been effectively used in a range of industries (Choi

et al. 2012). Other research also suggested that recommendation system may have an effect on how users and platforms interact, strengthening user loyalty (Xu, Peak and Prybutok 2015), influencing users online clicking and browsing habit (Besbes, Gur and Zeevi 2016) and influencing users' willingness to pay.

In this research we explore the complex world of product recommendation systems leveraging on customer online engagement data. There are two fundamental tasks involved in the recommendation process, the first step is to determine the users rating and relevance for previously undiscovered item. The prediction is centered on users' previous behaviour and their top-rated things are suggested to them as the second task. They are categorised into collaborative filtering, content-based, utility based, demographic based, knowledge based and hybrid-based as shown in **Figure1** below.

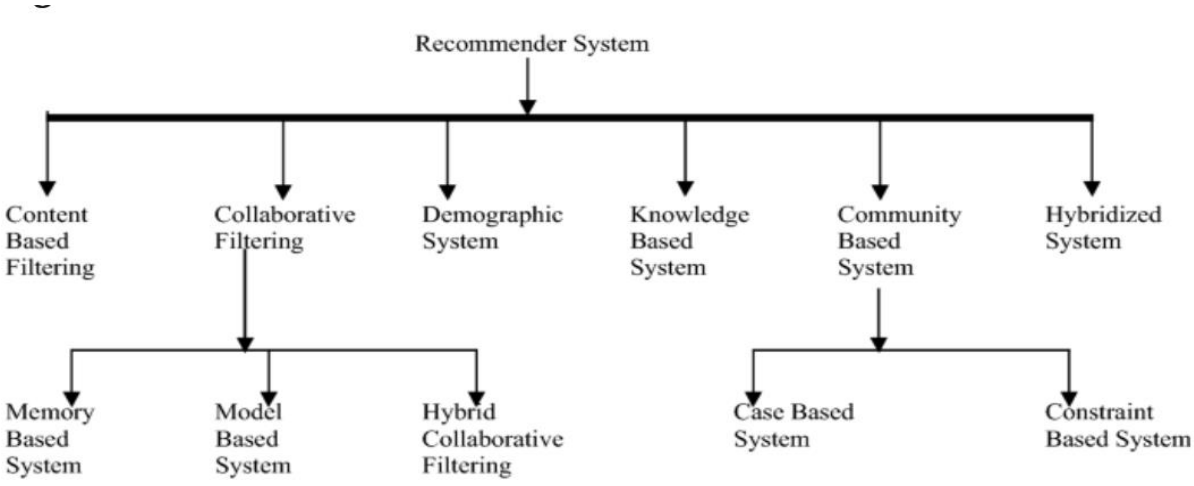


Figure 1 Recommendation system techniques

2.5.1 Collaborative Filtering Recommendation System

Suggestions are made to target customers based on the idea that other customer with similar tastes previously enjoyed them. By analysing the prior customer scores, you can determine how close two or more customers point of view are. Collaborative filtering algorithms have the ability to suggest or predict something new that few number of users will find appealing based

on past scores. Collaborative filtering also evaluates items user ratings either explicit or implicit from past data(Herlocker et al. 2004a). Database of user preferences are created and active users will be matched with neighbours who have similar purchasing behaviour as shown in **Figure 2**. Collaborative filtering system is further subdivided to Memory Based, Model Based and the Hybrid system as shown in **Figure 2**. The memory based can either be user-based filtering and item-based filtering. The user-based works by first identifying similar users to the target user and then obtain ratings which is then used to suggest recommendations. The item-based filtering system leverages on closeness and the similarity between items rather than users. It works by identifying an item and look for users who enjoyed it and locates other items the users will enjoy. The popular measure that is used to calculate similarities among users are Pearson correlation, cosine similarity, Spearman correlation and adjusted cosine similarity (Schafer et al. 2007).

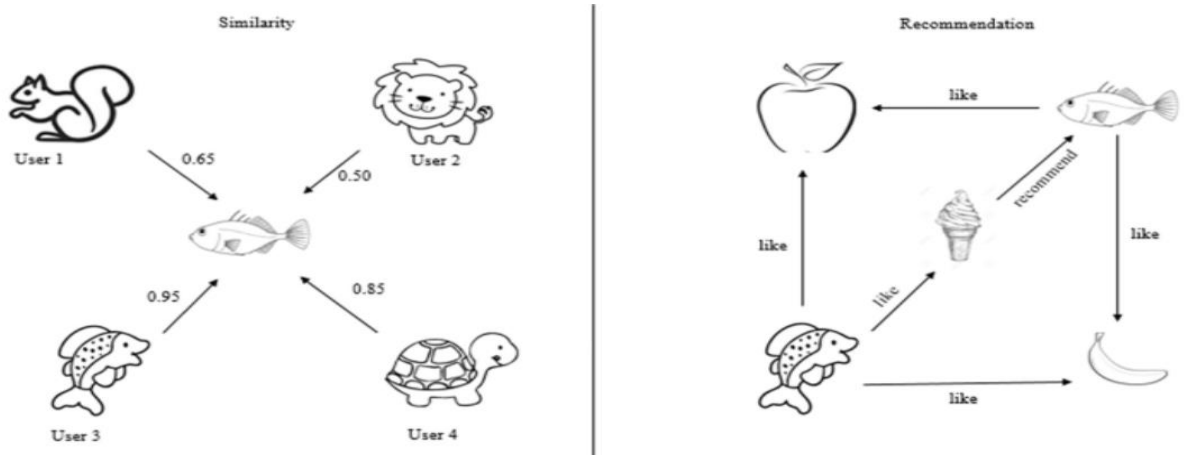


Figure 2 Collaborative filtering system (Sinha and Dhanalakshmi 2019)

2.5.2 Advantages of Collaborative Filtering

One main advantage of collaborative filtering techniques is that, it does not require the domain knowledge of the user to make predictions and its scalability does not require human intervention (Sinha and Dhanalakshmi 2019).

2.5.3 Disadvantages of Collaborative Filtering

It causes sparsity issues because majority of products and services on ecommerce sites do not have ratings. It is a challenge because until an item is rated by a user or demonstrates similarity to other products it cannot be recommended to any user(Sinha and Dhanalakshmi 2019).

2.5.4 Content-Based Recommendation System

The content-based approach creates a user profile in order to predict ratings for items yet to be seen. It mostly utilises tags, keywords and the utility of content-based filtering system is determined using functions such as cosine similarity metric. Content-based filtering is mostly used where the features values are extracted and not when the values must be manually entered. Content-based system simply analyses the items and user profile in order to provide suggestions. One major draw back of content-based system is that when no enough information is provided to identify products, the recommendations will not be accurate as it requires extensive domain knowledge (Gong and Cheng 2008).

2.5.5 Demographic System

Demographic correlation has been shown to enhance collaborative filtering methods in a number of quantitative research publications(Vozalis and Margaritis 2004). Demographic recommendation system provide recommendations by grouping users according to their demographic attributes. When the amount of item information is limited, demographic system comes in handy as it is able to address scalability and cold start problem. The system uses demographic features such as gender, age, language to make suggestions. One major advantage of this technique is that it is very precise in getting results from few analyses. Additionally, one disadvantage of this system is that due to general privacy and security issues it is impossible to obtain all user demographic attributes to be able to make recommendations. Another

challenge of this system is the difficulty in adjusting a customer's profile when their needs changes this is known as stability vs plasticity problem (Fayyaz et al. 2020).

2.5.6 Knowledge Based System.

This system makes use of knowledge about products and users to create a criterion for recommendations (Burke 1999). Knowledge based system does not depend on user ratings so it does not require large amount of data (Burke 2000). By analysing products that meets the user needs suggestions can be made to suite their preferences. One key advantage of the knowledge-based system is that, it does not need to gather information about a user because the suggestions do not depend on users tastes or preferences. Based on this unique feature of knowledge-based system is can be considered as a stand alone and can also complement other types of recommendation systems. One setback of the knowledge-based system is potential bottleneck in acquiring knowledge about users and products as the knowledge is acquired via rules and objects.

2.5.7 Hybrid System

The Hybrid combines two or more techniques to achieve better performance as shown in **Figure 3**. The main advantage is to eliminate the challenges of a single system. Some of the combination strategies are explained below.

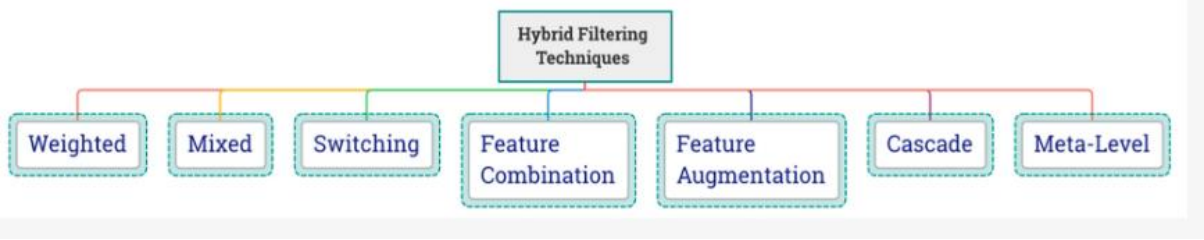


Figure 3 Hybrid System (Fayyaz et al. 2020)

2.5.7.1 Combination Strategies of the Hybrid System

The combination strategies of the hybrid recommendation system are discussed below.

2.5.7.2 Weighted

The output of many recommendation techniques is combined by a weighted-hybrid recommender. It uses a weighted average to combine the values from these approaches and determines the final score for each item. At first, the same weight is assigned to each approach. But as time goes on, the weights are changed according on how accurately each approach forecasts user ratings. Better-performing methods will have more weights. However, this model implicitly assumes that individual techniques' relative value is uniform among possible items, which is not always true(Kardan and Ebrahimi 2013a).

2.5.7.3 Mixed

A mixed hybrid approach makes sense when numerous recommendations are required at once. The mixed hybrid technique presents a consolidated list of its component recommendations side by side. The goal of this hybridization technique is not to attempt to combine recommenders' evidence. For this strategy, combining multiple independent lists is a difficult task. Conventional methods encompass merging based on the recommender's confidence or based on the expected rating (Kardan and Ebrahimi 2013b).

2.5.7.4 Switching

Among the constituents, one recommender is chosen by a switching technique. A different system could be selected for a different user or profile. For example, a different approach, such as the collaborative procedure, is tried if the content-based technique is unable to provide an accurate recommendation with high confidence. Not all of the problems that RSs face are avoided by this strategy (such as the ramp-up difficulty). The hybridization approach operates

on the assumption that a trustworthy standard exists for which to choose a switching option. The remaining undetermined elements of the left suggestion process are rendered irrelevant once the switching decision has been made (Burke 2000).

2.5.7.5 Feature Combination

Combining complementing elements from different techniques like collaborative-based recommendations into an algorithm designed to handle data using a different approach is known as feature combination (for example, content-based recommendation). By treating collaborative information as an extra feature data connected to each model and applying content-based approaches over this accumulated dataset, content-collaborative merging is accomplished. This method lessens the system's sensitivity to the quantity of users who rated an item by allowing it to take into account collective data without entirely depending on it (Herlocker et al. 2004b).

2.6 Recommendation Techniques

Data mining is a technique for obtaining useful insights by identifying patterns and correlations in large dataset (Lika, Kolomvatsos and Hadjiefthymiades 2014). In order to establish the client/visitor click stream matching data, it creates similar user groups based on the analysis of item information. Also, in order to meet the demands of every user it can suggest customise browsing options(Gündüz and Özsu 2003). The types of recommendation techniques are Text mining, KNN, Clustering, Matrix Factorization and Neural Network as shown in **Figure 4.**

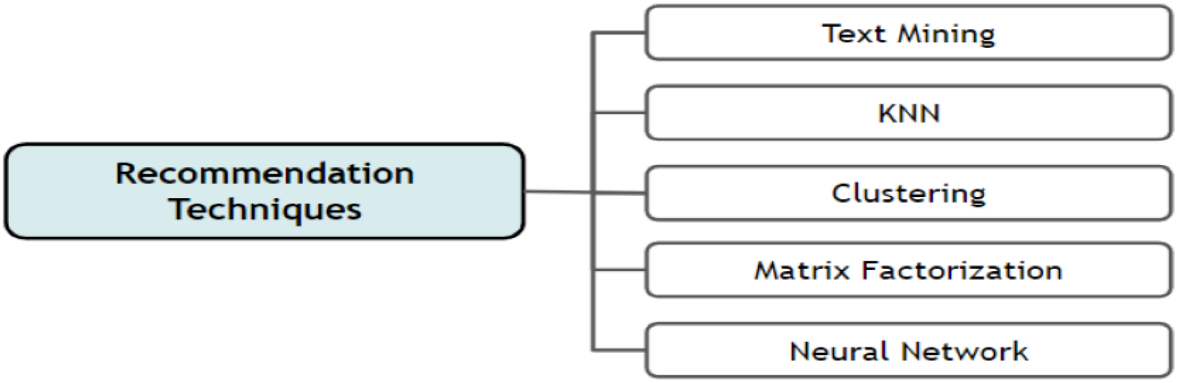


Figure 4 recommendation techniques

2.6.1 Types of recommendation Techniques

2.6.2 Text Mining

Text mining is a technique to discover useful information by retrieving text- related information from data. With the recent advancement in natural language processing technology key information can be derived from the corresponding text (Gómez-Pérez and Corcho 2002). When natural language processing is employed in some text analysis process, it is likely to analyse the texts using the occurrence so there are limitations to it use. So, in order to precisely get the meaning of the text the ontology(Noy and McGuinness 2001), which explains the common vocabulary of items and groups the meaning by building a text domain conceptual schema.

Additionally, by conducting a semantic analysis of the item of the item information in the Content Based System, text mining can be utilised to suggest similar items (Cantador et al. 2008). **Figure 5** shows a typical text mining process

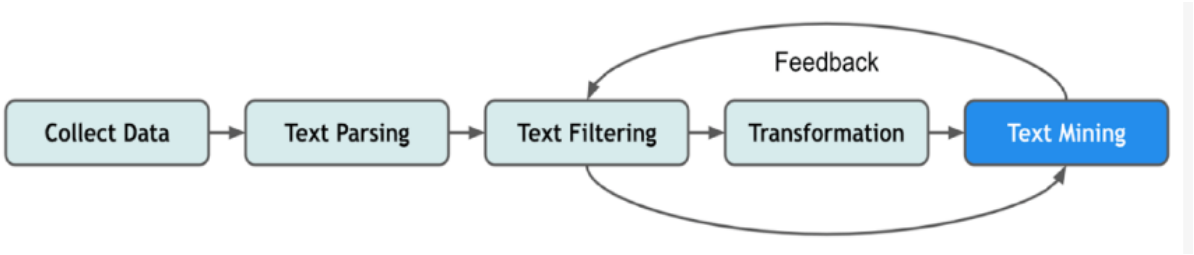


Figure 5 Text mining process

2.6.3 K-NN (K-Nearest Neighbour)

K-Nearest Neighbour (k-NN) is a machine learning algorithm that utilises the K nearest neighbour of the test and train tuples to group dataset. By using distance-based weighting we can classify dataset by analysing the similarities between items (Guo et al. 2003). The metrics used in k-NN to compare the similarities between data are the Euclidean distance, Cosine similarity and Pearson Correlation. For a recommendation system to successfully implement the k-NN algorithm, the users search pattern should first be categorised and then their future preferences can be forecasted. Also, it can be used to categorised products that are comparable to users' preferences after studying the patterns in the user activity data such as weblogs and click streams data.

Further more, (Jannach et al. 2015a) also published research suggesting that the k-NN algorithm-based recommendation model's performance was subpar. The model's performance is determined by K, and k-NN must choose a suitable value for it. This leads to inefficiencies that necessitate repeated tests and the issue of K bias. Furthermore, while evaluating data with a big input size, k-NN performs less well. Therefore, dimensionality reduction which turns the data into an understandable expression should be utilized when the magnitude of the input data is enormous. Reducing data without losing information is the primary objective of dimensionality reduction, and principal component analysis (PCA) and linear discriminant analysis (LDA) are frequently employed to achieve this.

2.6.4 Clustering

Clustering algorithm locates discrete categories or clusters to identify data and it used in recommendation due to its minimal redundancy and ambiguity (Shepitsen et al. 2008). K-means clustering is the commonly used clustering algorithm and its groups around the mean after determining the number of K clusters as shown in **Figure 6**. To determine the cluster centre, the recommendation system assigns data to the closest cluster using the similarities between all the data and then repeat all the calculation (Bilge and Polat 2013). Additionally, the K means clustering is prone to scalability problem when the number of clusters and users increases(Gan, Ma and Wu 2020). Clustering was extensively researched in recommendation systems in the tourism, education, and e-commerce domains and was primarily employed in the Collaborative Filtering recommendation model.

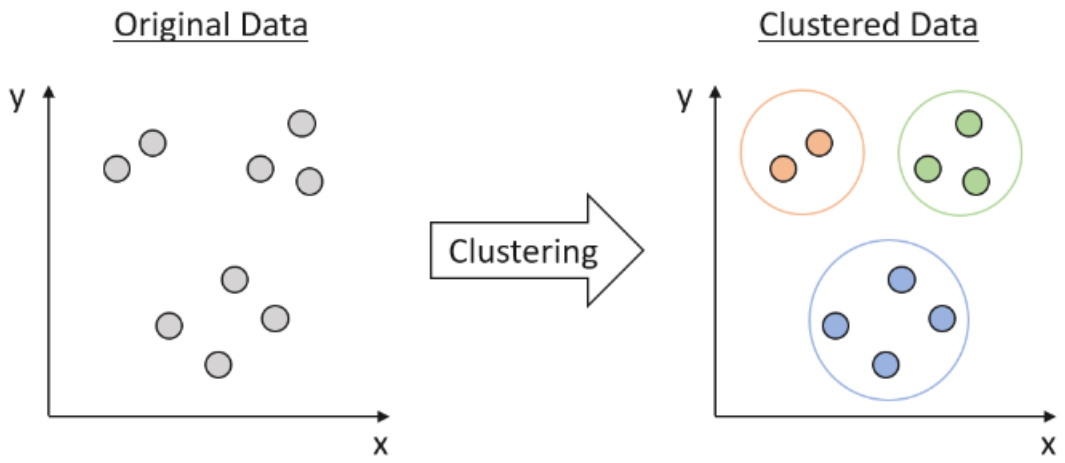


Figure 6

2.6.5 Matrix Factorisation

The Netflix Prize made the Matrix Factorization recommendation system well-known, and it particularly addresses the scarcity issue in collaborative filtering (Mnih and Salakhutdinov 2007). After determining elements from user evaluation data for things and storing them as vectors, matrix factorization is a technique for characterizing both items and user data (Koren,

Bell and Volinsky 2009). By storing the user's assessment data in the Rating Matrix, this technique aims to identify the dimension of the latent factor that conveys the user's information and preference (Kim et al. 2016). It provides good flexibility and scalability in particular because it may use explicit data that the user directly evaluates for a given item, as well as data from mouse movements and search patterns, to analyse users' preferences (Koren, Bell and Volinsky 2009). Singular value decomposition (SVD) is a technique that turns the user and their chosen items into a space of the same latent component by evaluating different item data (Koren 2008). The algorithm has the benefit of lowering computational overload, which raises the recommendation system's efficiency. Because it was feasible for several users to provide suggestions based on numerical evaluation of common items, it was challenging to distinguish user preferences with tag and like data in the SNS field in the present recommendation system (Ma et al. 2008).

2.7 Research Trends of Recommendation System Techniques

Figure 7 below analyse the research trends of techniques employed in recommendation systems. Frequency (number) of papers using recommendation techniques during the period investigated in this paper (2010 - 2020).

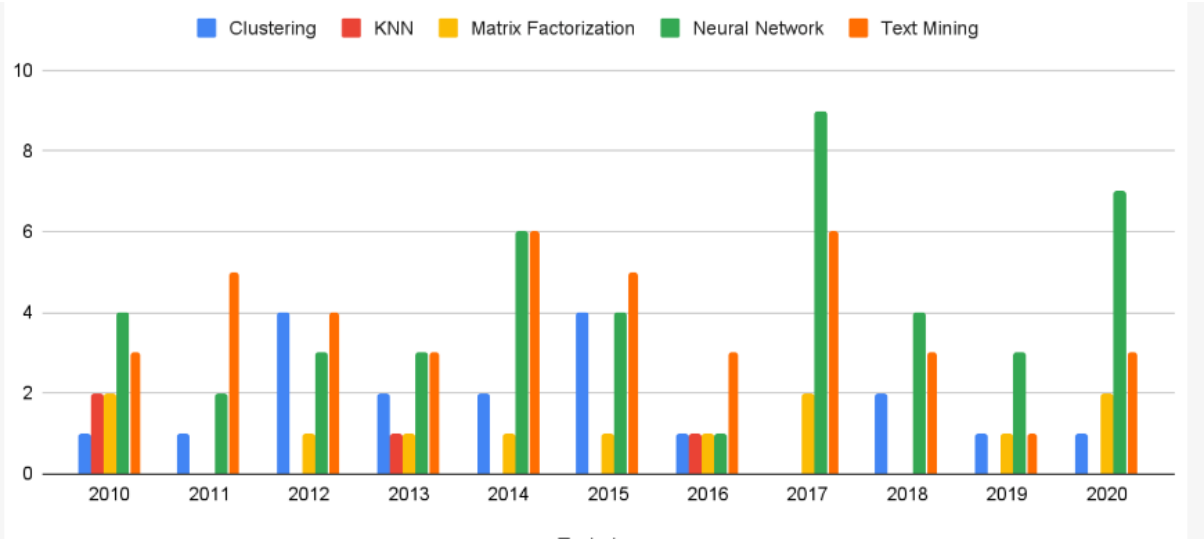


Figure 7 Trends in Recommendation (Ko et al. 2022)

Trend in recommendation technique papers by year (by period) during the period investigated in this paper (2010 - 2020). Text mining is the algorithm used to access the features of items chosen the user in a Collaborative filtering model such as Item-based filtering of the Content Based recommendation and the Hybrid System. This method is applicable to a variety of recommendation system model and in constant demand in areas that utilises a lot of text information such as academia, medical and tourism. From **Figure 7** it can be seen that text mining is widely used in recommendation system research.

Also, k-NN was the least utilised in recommendation system research after 2010, because of the difficult procedure involved in searching for the K values, biased in K and its inability to be used in big dataset (Jannach et al. 2015b).

Furthermore, with the Clustering technique it ultimate goal is to find user groups that are close to the users in the collaborative filtering model. The primary use of clustering in the Hybrid Recommendation model and Content-Based Filtering is in the analysis and clustering of location-based data similarity in the trip recommendation system sector. When analysing similar user groups or items and the assessment data is not numerical clustering is very helpful. In recent times, the usage of clustering as a recommendation technique is declining because users rate items using likes, star ratings or numerical data. From **Figure 7** it is shown that clustering is used consistently throughout the year.

Lastly, the Matrix Factorisation technique looks for possible indicators that explains the user's choice for a product or service it offers. MF is able to analyse numerical data as well data obtained from other sources such as mouse streams and external situation data. Because the computation is done by breaking down the matrix that contains the user and item evaluations, the calculation time can also be decreased. As a result, more research is now using MF than k-NN due to the addition of new k-NN challenges to the old ones. To develop a model that can calculate more than MF technology and produce results faster and more precisely than before,

however, a new recommendation system utilization technology is required as the number of organizations utilizing large-scale servers rises.

2.8 Real Applications of Recommendation Systems.

No single recommendation strategy works well for all users, most of websites have different recommendation approaches to offer relevant suggestions to end users.

In this section we present some popular ecommerce business that employs one or more variations of the recommendation systems technology in their website. For each website we give a summary of the recommendations provided, the type of technology used and the information gathered from users or customers.

2.8.1 Amazon

Our primary focus is on the recommendation systems of www.Amazon.com book section.

Customer who Purchased: This feature is located at the information page for each book in their catalogue. There are two unique recommendation lists. The first suggest books that customers who bought the chosen book frequently purchased. The second list also suggest authors whose books are regularly bought by readers who have already purchased the selected books.

Eyes: Customers submit request based on criteria such as title, author, subject ISBN and publishing dates and they receive email notifications anytime new books are uploaded to the Amazon.com inventory.

Amazon.com Delivers: This is a variation to the eyes feature; customers select from a list of particular genres such as biographies or books by a specific author. Occasionally the editors at Amazon.com send email updates to subscribers informing them about latest recommendations.

Book Matcher: Customers use a 5-point rating system to provide candid feedback on books they have read using this function. Users can ask for recommendations for novel they might enjoy after rating a sample of available books.

Customer Comments: This feature allows users to receive text recommendations based on the thoughts of other customers. A list of comments and reviews left by customers who have read the particular is found on the information page of each book and users can choose to take these suggestions into consideration while making a purchase(Schafer, Konstan and Riedl 1999).

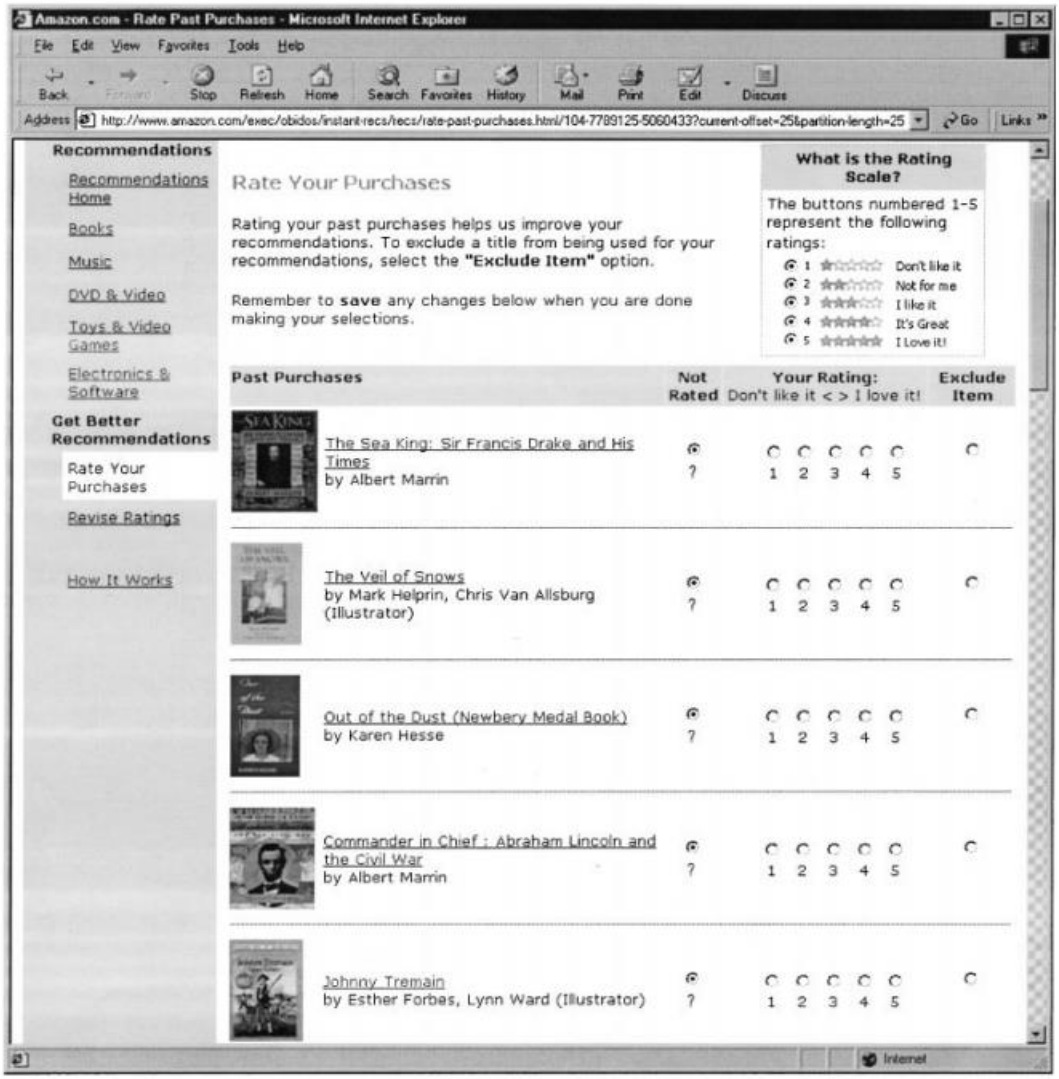


Figure 8 ratings page of amazon

The ratings page of amazon.com prompts informs customers to rate their recently purchased items as shown in **Figure 8**. These inputs are used by the recommendation engine to suggest similar items. Again, customers are encouraging to rate items and in return get more useful suggestions (Schafer, Konstan and Riedl 2001).

2.8.2 Netflix

The homepage of a Netflix subscriber is where all the Netflix personalization commences and it appears on any device after the user logs in. The movies on the page are organised into groups and arranged in horizontal rows to form two dimensional grids. The title of each rows videos shows the relationship between the videos in that row. **Figure 9**, shows the best estimate of a video a user is likely to watch. In Netflix a user is every member of the household and the recommendation handles different people within the household with different taste. In the top 10 rows, a user in the family is likely to discover item for mom, dad and children. Additionally, for a single person household the recommendations are tailored to meet the persons interests and mood.



Figure 9 Example of Netflix top 10 rows ((Molina and Bhulai 2018)

Another key element of Netflix personalisation is awareness, users are made to understand how changes are being made to suite their preferences. This further improves recommendation by encouraging users to give feedback which results in better recommendations. In Netflix a video is recommended to a user because it matches the information that is kept about the user for example feedback ratings, viewing history and even a friend's recommendation.

Furthermore, Netflix has a social feature that allows users to connect through Facebook to find friends who are also Netflix users. This allows Netflix to generate rows of recommendations based on social circle. Netflix also employs the concept of similarity as an important aspect of its personalisation. A row of "adhoc genres" based on their similarity to the videos a user has interacted with which is saved with a label "Because you watched" as shown in **Figure 10**. Also, a list of related videos a subscriber may be interested in the information page.

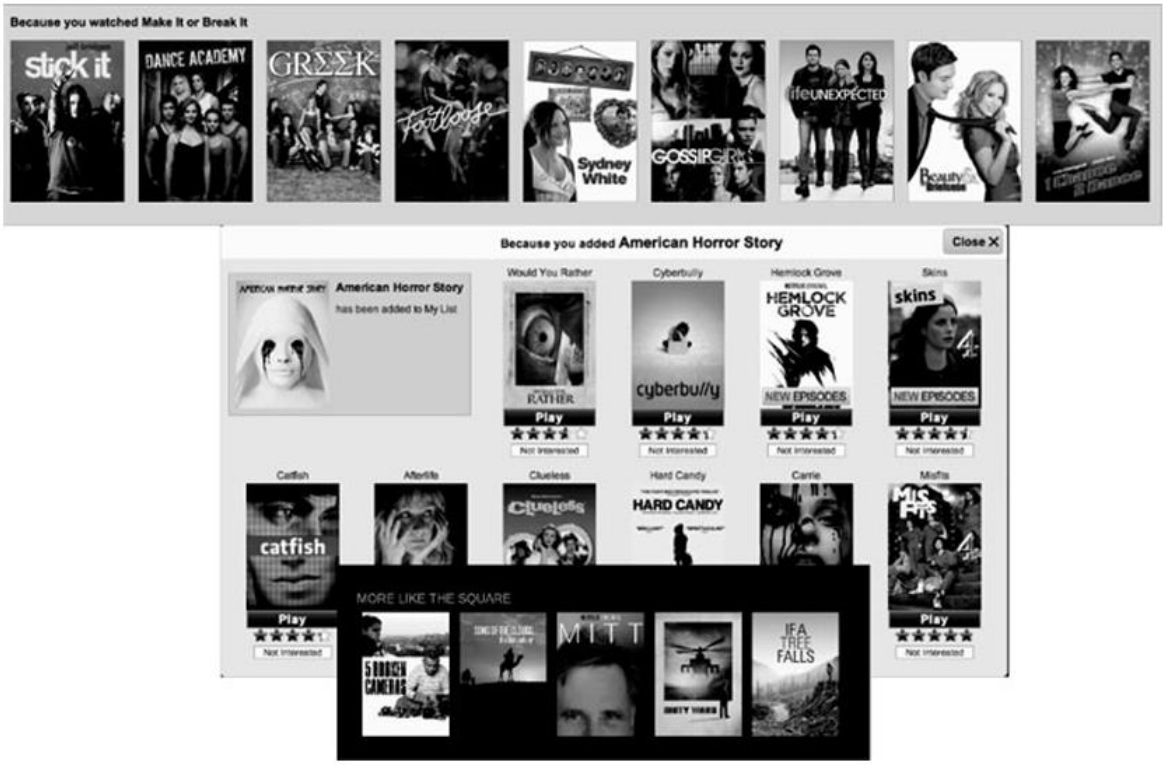


Figure 10 Similarity used to provide recommendations (Molina and Bhulai 2018)

2.9 Evaluation Metrics

In evaluating a recommendation system, the dataset is taken as input and then split into test and training dataset. The train data is used as input for the recommendation system which suggest recommendations as output. The out put is analysed against the test data and measured using evaluation metrics as shown in **Figure 11**.

Recommendation system evaluation requires establishing the features that makes an excellent system and how it can be measured. The metrics for measuring its performance are precision, recall, accuracy, F-measure, ROC curve and Mean Average Precision (MAP).

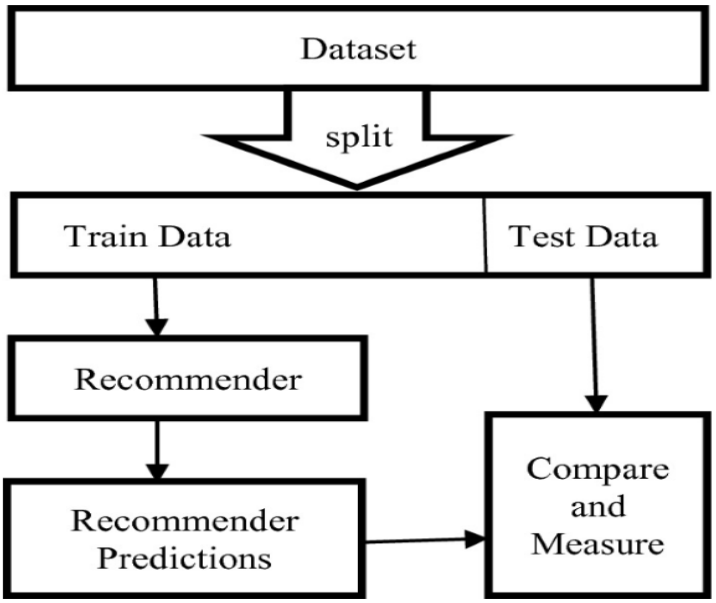


Figure 11

2.9.1 Recall and Precision

Information retrieval (IR) is centered on suggesting relevant items from a pool and works the same way as a recommendation system offering relevant items from a pool. The IR is considered to be sufficient source of tools for recommendation system example is measurement metrics. “Precision and Recall” are the two key metrics. Recall is the proportion of key items to the total number of objects that should be suggested to a user, on the other hand Precision is the percentage of key items among the total number of objects recommended to the user. Any item that is of benefit to the user is considered as a key item. In order to be able to determine the Precision and Recall a confusion matrix has to be computed.

	Successful Recommendation	Not a Successful Recommendation
Recommended	a	b
Not Recommended	c	d

Figure 12 A confusion matrix for a recommendation system (Fayyaz et al. 2020)

The confusion matrix in **Figure 12** shows four possible outcomes, where "a" represents the number of items that has been successfully recommended, "b" represents items that are unsuccessfully recommended, "c" also shows disqualified items that are successfully recommended and finally "d" shows disqualified items that were not recommended.

A good recommendation system tries to take advantage of both metrics in its analysis for example it can recommend many products/items to the user, obtaining maximum coverage. The Precision would still be as low as the ratio of useful products/items in the pool (Fayyaz et al. 2020).

2.9.2 Accuracy

Choosing a recommendation system based on evaluation metric is a sophisticated choice that must be made based on the needs of the company. Predicted ratings are usually employed as a system for evaluation metric. Accessing the accuracy of recommendation system is not an easy task because there is no clear way to tell whether a recommendation is precise or not (Vaishnavi, Jayanthi and Karthik 2013). To access the accuracy of recommendation system one must look for minimal prediction errors in order to evaluate recommendation accuracy. For example, suppose we submit 80% of a user purchase history dataset to a RS and ask to predict the rest. In that case, we can calculate the system's accuracy based on true recommendations, as shown in **Figure 13** below.

$$Accuracy = \frac{Number\ of\ successful\ recomendations}{Total\ number\ of\ recommendations}$$

Figure 13

2.9.3 ROC Curve

Receiver operating characteristic (ROC) analysis can replace the precision/recall. A precision versus recall curve as shown in **Figure 14** below. The higher the precision is, the lower the recall points. The goal of ROC analysis is to obtain the key items without retrieving the irrelevant ones (Gunawardana and Shani 2009). This is achieved by having the fallout minimum, also known as the false positive rate, and the recall maximized, also known as the genuine positive rate. When the threshold is altered, ROC curves are used to graphically represent the trade-off between recall and precision, which aids in the classification of an item as "to be recommended" or "not to be recommended" (Del Olmo and Gaudioso 2008). Similar optimizations are made to the recall, accuracy, and ROC curves.

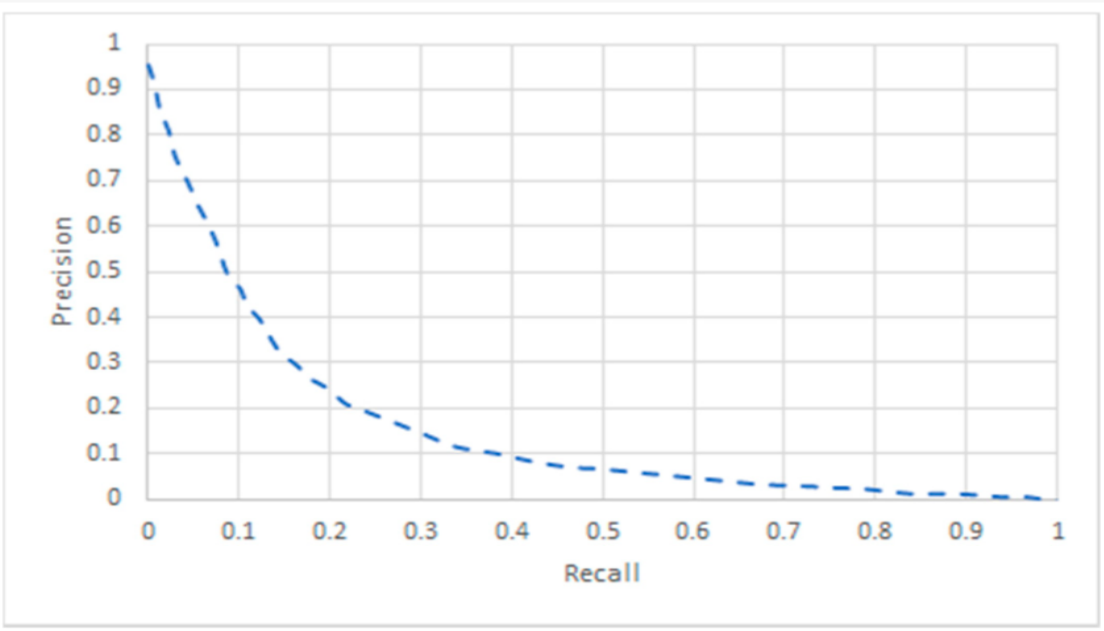


Figure 14

2.9.4 Recall

Another metric that is derived from recall and precision is the F-measure, which displays the characteristics of both measures. Since accuracy and recall provide different information that might complement each other when combined, F-measures may be a more helpful statistic than precision and recall alone. The metric with the highest level of excellence will be reflected in the F-measure. F-measure is the number of tests required to identify the first failure from a probability standpoint (Hanley and McNeil 1982). The value of Fβ assigns a greater weight to one statistic than the other by changing β. However, the consistent mean of precision and recall (F1), where β = ½, is the most widely used F-measure. Remember that the F-measure has a maximum value of 1, which indicates that all predictions are accurate recommendations. Understanding ranked retrieval is helpful. In this case, Recall@K is the ratio of relevant things in the top-k, and Precision@K is the ratio of the top-k relevant items. The premise behind ranked retrieval is that the user will look at just the top k results. Given this, demonstrates that the increasing recall value will lead the F-measure to rise along with the value of k. The computations of the Fβ and F1 are displayed in **Figure 15**.

$$F_\beta = \frac{precision \times recall}{(1-\beta) \times precision + \beta \times recall}$$

$$F_1 = \frac{2 \times precision \times recall}{precision + recall}$$

Figure 15

2.9.5 Mean Average Precision (MAP)

The average precision asks which fractions of recommendations are good. It treats all errors equally and utilises fixed sizes. For each relevant item in the recommendation list the precision

through that item is computed and the average sub-list of precisions is also calculated (Vineela et al. 2021).

2.10 Future Directions in Recommendation System

In this paper we aim at developing a recommendation model using ecommerce data, we first highlighted the general overview of recommendation systems, phases of recommendation system and recommendation systems in ecommerce. Additionally, accessed the various types of recommendation system such as collaborative filtering, content based, utility-based, demographic-based, hybrid-based and knowledge-based. We also explored the different combination strategies of the hybrid-based systems which is sub-divided into weighted, mixed, switch and feature combination. Additionally, we looked at recommendation techniques in ecommerce such as Text mining, KNN, Clustering and Matrix factorisation and the research trends of recommendation system techniques.

Further more, we presented a real application of recommendation system using Netflix and Amazon. For each website we give a summary of the recommendations provided, the type of technology used and the information gathered from users or customers. The following conclusions were made for future directions that hold significant potential for enhancing in ecommerce recommendation system:

Conversational Recommendation Systems: Building chatbots and virtual assistants that can engage users in natural language dialogue to provide personalised recommendations.

Context-Aware Recommendations: Leveraging contextual information such as time, location, and device to provide more relevant recommendations.

Reinforcement Learning: Utilizing reinforcement learning to continuously adapt recommendations based on user interactions and feedback in real-time.

Chapter 3

3.1 Methodology

This chapter covers the methodology used to develop the recommendation model and how it is adjusted to meet the project goals and objectives.

3.2 Data Understanding

The datasets were obtained from Mason Greenwood ecommerce data from Havard Dataverse. The are two sets of data we used for this research customer data and purchase data. The customer data has customer interaction information such as Email, Address, Avartar, Time on App, Time on Website, Length of Membership and Yearly Amount Spent as shown in the **Figure 16** below.

	\tEmail	Address	Avatar	Time on App	Time on Website	Length of Membership	Yearly Amount Spent
0	mstephenson@fernandez.com	835 Frank TunnelWrightmouth, MI 82180-9605	Violet	12.66	39.58	4.08	587.95
1	hduke@hotmail.com	4547 Archer CommonDiazchester, CA 06566-8576	DarkGreen	11.11	37.27	2.66	392.20
2	pallen@yahoo.com	24645 Valerie Unions Suite 582Cobbborough, DC ...	Bisque	11.33	37.11	4.10	487.55
3	riverarebecca@gmail.com	1414 David ThroughwayPort Jason, OH 22070-1220	SaddleBrown	13.72	36.72	3.12	581.85
4	mstephens@davidson-herman.com	14023 Rodriguez PassagePort Jacobville, PR 372...	MediumAquaMarine	12.80	37.54	4.45	599.41

Figure 16 customer data

The purchase dataset also has information on customers purchase history such as Item purchased, Category, Size Colour as shown in **Figure 17** below. Our recommendation model leverages on the customer interaction information and the purchase history from the two datasets. Below is a brief description of all the engagement features that will be used in our model.

	Email	Item Purchased	Category	Purchase Amount (USD)	Size	Color	Season	Payment Method
0	mstephenson@fernandez.com	Blouse	Clothing	53	L	Gray	Winter	Venmo
1	hduke@hotmail.com	Sweater	Clothing	64	L	Maroon	Winter	Cash
2	pallen@yahoo.com	Jeans	Clothing	73	S	Maroon	Spring	Credit Card
3	riverarebecca@gmail.com	Sandals	Footwear	90	M	Maroon	Spring	PayPal
4	mstephens@davidson-herman.com	Blouse	Clothing	49	M	Turquoise	Spring	PayPal

Figure 17 purchase data

Customer identification (Email)

The email is the unique identifier for each customer and its very key when the ecommerce business want to send personalised marketing and communication. It relies heavily on direct digital marketing and customer relationship management (CRM).

Geographical Insights (Address)

Customer address also allows the business to tailor its offering to customers based on location. This ecommerce platforms deals with physical goods and utilises the customer address for delivery or shipping.

Online interaction (Time on App and Time on Website)

Time on App and Time on Website provide insights into customer online interaction behaviour across different platforms. This helps the ecommerce business to understand customer preferred shopping channels, optimise user experience and increase customer retention.

Financial Performance (Yearly Amount Spent)

The Yearly Amount Spent provides insights into customers annual spending crucial for segmenting the customers based on their purchase value that is identifying high value customer and designing loyalty programs.

Customer Loyalty and Retention (Length of Membership)

Length of Membership measures the loyalty and effectives of the ecommerce retention strategies. The business offers membership such as subscription boxes, premium services and loyalty programs. This metric allows to track customer satisfaction and loyalty.

3.3 Data Preparation

This stage details how the datasets were cleaned and transformed for analysis using python. Python was selected because of its rich libraries specifically designed for machine learning and data science.

3.3.1 Cleaning and normalising the numerical features

The data set had two columns the “Address” and “Avatar” that were not relevant to our recommendation module and were dropped as shown in **Figure 18**. Customer email is the unique identifier for all customers, we used python to check for duplicates and also remove white spaces in the email column to make sure it can serve as the unique customer id as shown in **Figure 18**. Additionally, all the numerical feature (Time on App, Time on Website and Yearly Amount Spent) were all normalised as shown in **Figure 19**. Different features in the dataset had different scales for example Time on App is measured in minutes and Yearly Amount Spent range from hundred to thousands. This was done to ensure that each feature

contributes equally to the module and also to reduce numerical instability as large numerical values can suffer from precision errors.

	\tEmail	Address	Avatar	Time on App	Time on Website	Length of Membership	Yearly Amount Spent
0	mstephenson@fernandez.com	835 Frank TunnelWrightmouth, MI 82180-9605	Violet	12.66	39.58	4.08	587.95
1	hduke@hotmail.com	4547 Archer CommonDiazchester, CA 06566-8576	DarkGreen	11.11	37.27	2.66	392.20
2	pallen@yahoo.com	24645 Valerie Unions Suite 582Cobbborough, DC ...	Bisque	11.33	37.11	4.10	487.55
3	riverarebecca@gmail.com	1414 David ThroughwayPort Jason, OH 22070-1220	SaddleBrown	13.72	36.72	3.12	581.85
4	mstephens@davidson-herman.com	14023 Rodriguez PassagePort Jacobville, PR 372...	MediumAquaMarine	12.80	37.54	4.45	599.41

Figure 18

	Email	Time on App	Time on Website	Length of Membership	Yearly Amount Spent
0	mstephenson@fernandez.com	0.626888	0.929508	0.572932	0.651037
1	hduke@hotmail.com	0.392749	0.550820	0.359398	0.266346
2	pallen@yahoo.com	0.425982	0.524590	0.575940	0.453729
3	riverarebecca@gmail.com	0.787009	0.460656	0.428571	0.639049
4	mstephens@davidson-herman.com	0.648036	0.595082	0.628571	0.673558
...	...	...	...	...	...

Figure 19 normalising all numerical features

3.4 Model Selection

In this phase, supervised machine learning algorithm was implemented to create a model and then used to create a recommendation engine. Following the literature recommendations the Memory-Based Collaborative Filtering System was adopted. This is because collaborative filtering system can be used to predict the preference of a user by collecting preferences from many other users. The memory-based approach was also used because it makes use of the entire user-item interaction to make predictions. In our dataset we wanted to utilise all the customer online interaction features and purchase history to build our model, this will give us to a holistic

view of each customer. The memory-based approach is further subdivided into user-based system, this method focuses on finding users who have similar preferences to the target user and then recommending items those similar users have liked or purchased. Based on our datasets, our project objective and the reasons outlined above the memory-based collaborative filtering system was chosen for our model.

Additionally, to be able to identify similarities among the users we employed the k-NN algorithm. It leverages the entire dataset to identify user-item interaction to make recommendation. This is done by finding most similar users based on their distance metric like Euclidean distance. The k-NN algorithm was chosen over the other popular algorithms like Neural Networks (deep learning) and Matrix Factorisation because of its quick calculation times as it requires less computation which is in line with our ethical consideration for this project. Again, it was considered because of the nature of our datasets and it quicker predictiveness.

3.5 Legal, Ethical and Professional issues

This section examines the possible issues of undertaking this research project.

3.5.1 Legal Considerations - Data Usage and Privacy.

In any research that involves data, steps must be taken to ensure that its use is in compliance to General Data Protection Regulation (GDPR) standards and as such this research should adhere to its principles. One major consideration is the choice to use machine learning instead of less intensive solution which requires less computational power and lower energy consumption (Floridi 2016). In addressing this we implemented a resource efficient model. That is an optimal k value was selected for our k-NN which provides a good balance between computational efficiency and accuracy. Additionally, steps were taken to ensure that the users

remain anonymous and the data involved is only used for academic purpose under academic supervision.

3.5.2 Ethical and Professional issues

In undertaking this project one major ethical concern is the extent of influence, the recommendation system filters and retrieve just the top occurring items among the neighbors as final recommendation. However, the other options are not hidden from the user but less visible. Even though users can still access information about these available options but they are likely to just consider the top recommendation. According to (Collins et al. 2018) the tendency for users of recommendation systems to engage more frequently with things at the top of a list than lower items at lower positions in the list regardless of the items actual relevance is called “position bias” and affects users of recommendation systems. These is a challenge because if the recommendations are being manipulated or inaccurate users are being pushed toward options that do not reflect their choice and are distracted from their original intension.

Additionally, another ethical concern is the likelihood of recommendation system to limit users’ exposure to narrow selection of items and eventually leading them to miss important content. This has prompted scholars to question whether less important information should still be included in initial recommendation a concept know as “Serendipity by design” (Reviglio 2017). To address these potential ethical issues all the customer interactions feature even the less relevant once were considered in our recommendation model, this is to encourage serendipity. This will help users to discover diverse perspective and reduce narrow or biased recommendation.

3.6 Chapter Summary

This chapter outlined the project's methodology and provided an overview of dataset, the data cleaning and normalisation of numerical features and the ethical considerations. The project's implementation will be covered in full in the upcoming chapter.

Chapter 4

4.1 Implementation

4.2 Visualisation and Statistical data analysis in Python

After normalising the numerical features and dropping the irrelevant columns, the actual data set needed for the analysis was created. An initial exploratory data analysis was conducted with python using functions from Pandas, NumPy, Seaborn and Matplotlib libraries. The code used can be referred to in the appendix. Now we try to understand the distribution using scatter plot and relationship between all the numerical features that will be used to build our model using covariance and correlation. Also, we analyse high spenders using statistics like mean, median and mode. This will give us deeper understanding of customer behaviour to enable the business make informed decisions.

A scatter plot was first plotted to establish the relationship between Time on App and the Yearly Amount Spent as shown in **Figure 21**. The relationship has a positive covariance value of 39.382, that is a higher time on the app translates into high yearly amount spent. This could be that the ecommerce site uses a well-designed user-friendly app interface that encourages the users to browse more and shop longer. Again, push notification on the app and smooth check out process such as integrating “Buy Now, Pay Later” services like Klarna and Clear pay on the app can make customers make repeat purchase leading to higher yearly amount spent.

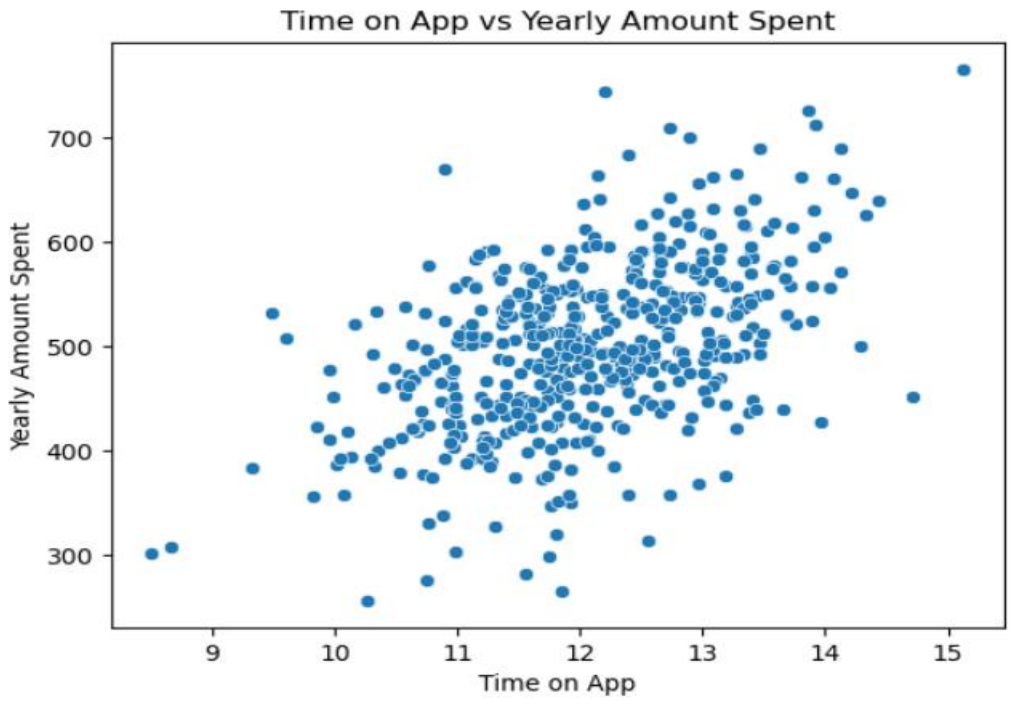


Figure 20 Time on App vs Yearly Amount Spent

Again, a scatter plot for Time on Website against Yearly Amount Spent as shown in **Figure 22** below was plotted to establish the relationship between the two. Surprisingly the relationship has a negative covariance value of - 0.208. This means that Time on Website is inversely proportional to the Yearly Amount customer spends. That is some users just browse the website without making significant purchases either they are indecisive, looking for deals or just exploring options without any a clear intention to buy. Another key factor that can cause this is that users might be overwhelmed with a number of choices leading to decision fatigue, where they spend more time but making limited purchases. This is where our recommendation system comes in handy to offer relevant recommendations to such users.

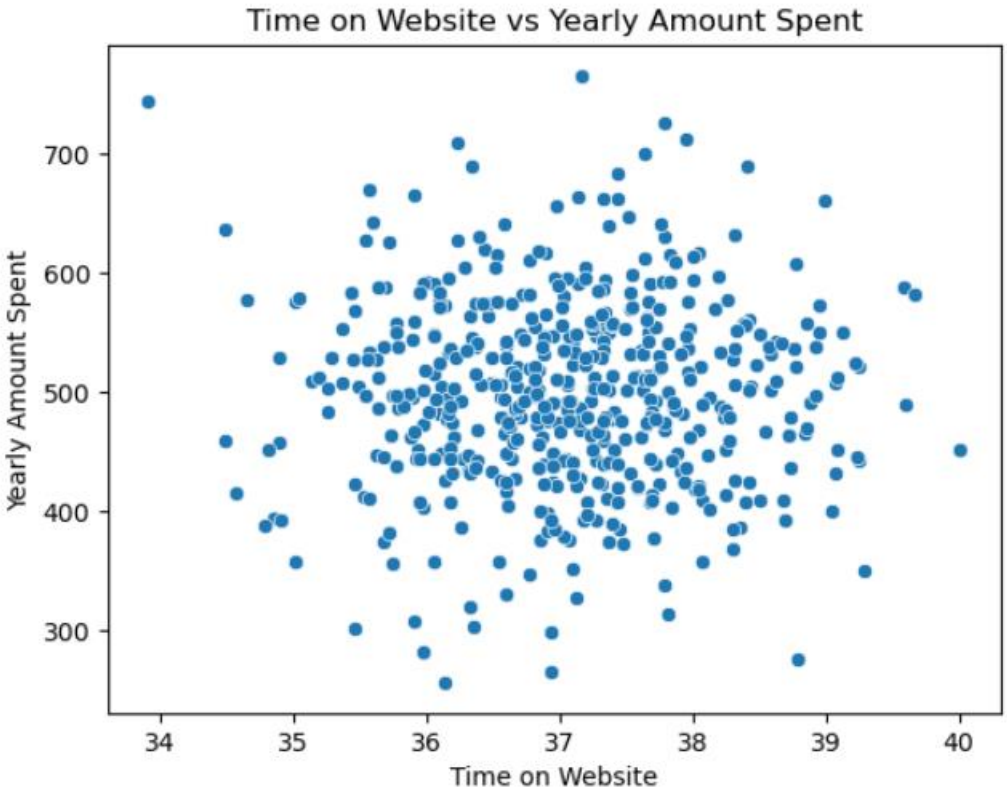


Figure 21 Time on Website vs Yearly Amount Spent

Also, Length of Membership was plotted against Yearly Amount Spent as shown in **Figure 23** below. The relationship has a positive covariance value of 64.123. This shows that the loyal customers tend to spend more. This is can be attributed to the ecommerce success in deploying loyalty programmes that offer rewards and discounts that encourages them to spend more. Also, longer members represent longer higher lifetime value to the business.



Figure 22 Length of Membership vs Yearly Amount Spent

Further more, a correlation heatmap was also plotted to establish the relationship between all the numerical features as shown in **Figure 24**. The values ranged from -1 to 1, showing the linear relationship between the numerical features. Length of Membership and Yearly Amount Spent has the strongest positive linear relationship with a value of 0.82 which shows that customers with longer membership tend to spend more annually. Alternatively, Time on Website and Yearly Amount Spent has a very weak negative linear relationship which also means that customers who utilise the website more often do not spend more annually.

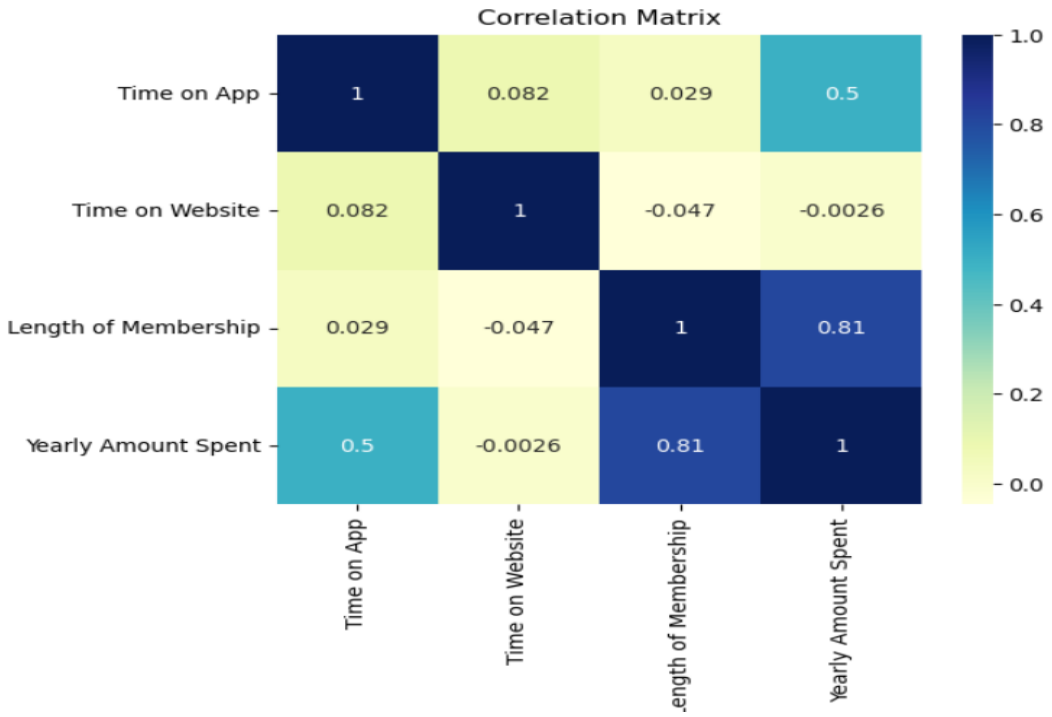


Figure 23 Correlation heatmap

Again, we utilised summary statistics to identify purchasing patterns. From **Figure 25** below the median Yearly amount spent is 498.89, also it can be observed that all the top spenders (Above 90th Percentile) tend to utilise the website more for their transactions. The time they spend on the website is more than twice the time they spend on the app and this is very useful insight for the ecommerce business on how to attract and retain the high-end customers.

```
Mean Yearly Amount Spent: 499.31424
Median Yearly Amount Spent: 498.89
Mode Yearly Amount Spent: 357.86
Maximum Yearly Amount Spent: 765.52
Minimum Yearly Amount Spent: 256.67
90th Percentile Yearly Amount Spent: 593.236
Highest Spenders (Above 90th Percentile):
                                Email  Time on App  Time on Website  \
4        mstephens@davidson-herman.com        12.80            37.54
5               alvareznancy@lucas.biz        12.03            34.48
19              samuel46@love-west.net        13.99            37.19
24              youngbarbara@yahoo.com        12.89            37.64
26           ijones@schaefer-carr.net         12.78            36.43
46             gonzaleskatie@gmail.com        13.15            37.34
50                 william82@gmail.com        13.86            37.78
```

Figure 24 High spenders

Similarly, the low spenders that is (below the 10th Percentile) tend to also spend more time on the website as compared to the app as shown in **Figure 26**.

```
Lower Spenders (Below 10th Percentile):
                              Email  Time on App  Time on Website  \
1                   hduke@hotmail.com        11.11            37.27
36                    enash@gmail.com        11.76            36.77
52              josephgould@west.info        10.32            37.45
83   shafferdaniel@murphy-harris.info        10.88            37.78
101                  mark52@yahoo.com        11.28            37.39
107              amandadean@gmail.com        10.72            37.71
143               kaitlyn78@chang.com        11.92            39.29
161          stevenking@patterson.com        12.40            35.01
173           fischermonica@gmail.com        11.75            36.94
```

Figure 25 Low spenders

In building our model we utilised the two ecommerce datasets, the customer data and the purchase data. We then merge the two datasets to create a unified dataset on the “Email” which serve as the unique identifier. This unified dataset contains the engagement features of each customer along with their purchase history as shown in **Figure 27**.

	Email	Time on App	Time on Website	Length of Membership	Yearly Amount Spent	Item Purchased	Category	Purchase Amount (USD)	Size	Color	Season	Payment Method
0	mstephenson@fernandez.com	12.66	39.58	4.08	587.95	Blouse	Clothing	53	L	Gray	Winter	Venmo
1	mstephenson@fernandez.com	12.66	39.58	4.08	587.95	T-shirt	Clothing	31	L	Purple	Fall	Venmo
2	mstephenson@fernandez.com	12.66	39.58	4.08	587.95	Socks	Clothing	46	M	Red	Winter	Bank Transfer
3	mstephenson@fernandez.com	12.66	39.58	4.08	587.95	Jeans	Clothing	69	L	Maroon	Summer	Debit Card
4	mstephenson@fernandez.com	12.66	39.58	4.08	587.95	Belt	Accessories	27	M	Lavender	Winter	Cash

Figure 26 unified dataset

4.4 Determining the Best Value of K

In using our k-NN algorithm we need to get the k value that is the number of nearest neighbours we want our algorithm to consider in making recommendations. This is very crucial because it directly impacts our model performance, a very low k value will make the model highly sensitive to noise or outliers in the training data. A higher k is likely to oversimplify the data leading to underfitting. Our goal here is to find the optimal k value, that maximises the accuracy of the model during cross-validation. Cross validation is used to determine the various values of k. Each value of k is plotted against its accuracy to determine the one that gives the highest cross validation accuracy. From **Figure 28** below the best k value is 5 (highest value).

Our data is then split into training and testing sets with 70% used for training and the remaining 30% for testing and to ensure reproducibility our random state is set to 40. The 70/30 split provides a good balance allowing the model to learn effectively and still reserving enough data to access it ability to make generalise recommendations.

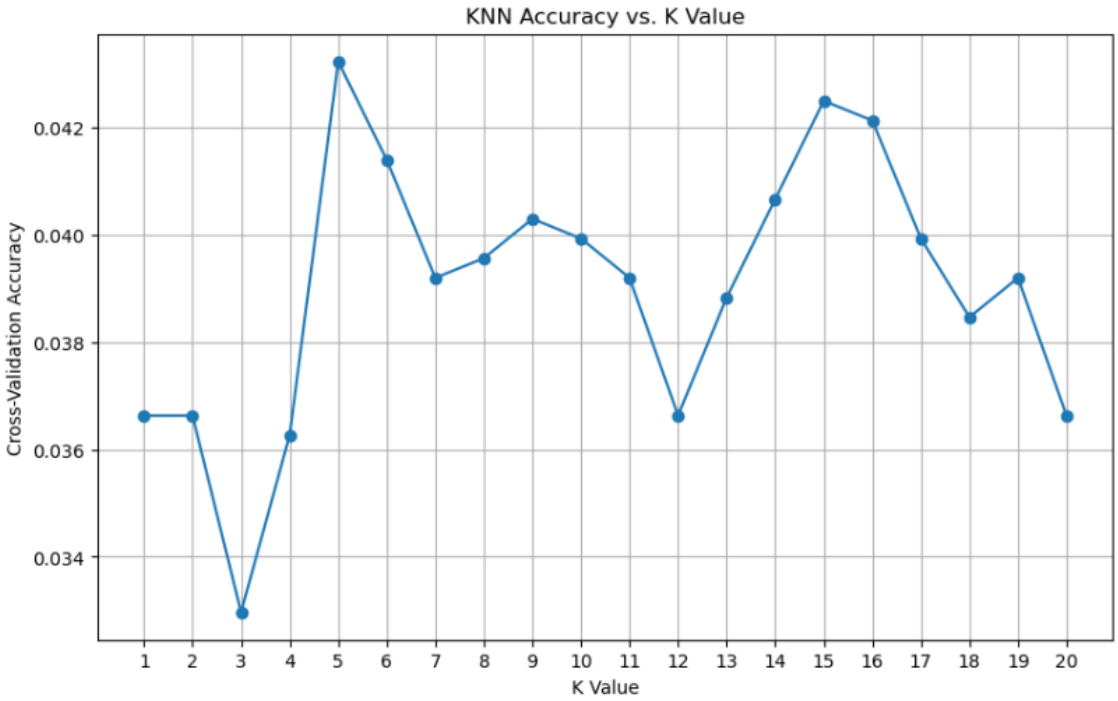


Figure 27 k value using cross validation

4.5 Recommendation Function using collaborative filtering and k=5

Our recommendation function is designed to recommend items to a user based on the items purchased by another user. The steps below are used to build our model using the best k=5.

Step 1

The email is normalised and converted to lowercase; this is to ensure consistency on how emails are compared. By using the email, the function first checks if the user exists in the unified dataset, if not it returns a message indicating that the user is not found to prevent further processing of an unknown user.

Step 2

The function then extracts the data of the user and then select the relevant features (Time on App, Time on Website, Yearly Amount Spent and Length of Membership) which will be used

to identify similar uses. It then utilises “StandardScaler” to ensure they are on the same scale to make them comparable.

Step 3

The k-NN algorithm is created using “NearestNeighbors”, the k value is set to 5 as the number of nearest neighbors it should consider.

Step4

The “kneighbors” is used to find the k nearest neighbors to the target user based on the Euclidean distance between the user’s scaled features and the features of other users in the dataset.

Step5

Using the indices of the nearest neighbors, the function retrieves the corresponding emails from the unified dataset. These emails represent users who are most similar to the target user.

Step6

The function retrieves items purchased by the target user and the items purchased by the neighbours and. filters out any items the target user has already bought ensuring that only new items are recommended.

Step7

Finally, the function counts how often each item appears in the neighbors’ purchases, it then returns top items as final recommendations based on their frequency among the neighbors

Step8

Finally, a recommendation engine for the model is developed based on the function, it takes a user email as an input and then suggest items excluding the items already purchased by the user as recommendations.

4.6 Conclusions

This chapter summarized the steps taken during the implementation phase of this project. The analysis included data understanding as well as understanding the distribution of the data and how the numerical feature relate to each other.

Also, the recommendation model was developed using memory based collaborating filtering system that utilises k-NN algorithm to recommend items to users.

Chapter 5

5.1 Results and Discussion.

In this chapter we discuss and evaluate the outcome of implementing a recommendation model that employs memory-based collaborative filtering model on our ecommerce dataset. The main objective is to provide a personalised recommendations to a target user based on their similarity to other users.

5.2 Recommendation Engine

To effectively test our recommendation model, we built a recommendation engine, that takes the email of a target user as an input to find the nearest neighbours. It then retrieves the items purchased by the target user and also the items purchase by the neighbors. The function compares the two and recommends the items in the neighbors list that are yet to be purchased by the target user as recommendation as shown in **Figure 29** below.

```
# Example usage
email_to_recommend = 'lisagutierrez@johnson-herring.net'
recommendations = recommend_knn(email_to_recommend, customer_purchase, X_scaled, k=5)
print("Recommendations for", email_to_recommend, ":", recommendations)
```

```
Recommendations for lisagutierrez@johnson-herring.net : ['Belt', 'Jeans', 'Blouse', 'Jewelry', 'Shoes']
```

Figure 28 recommendation engine

A random user "**lisagutierrez@johnson-herring.net"** from the dataset was selected for generating recommendations. The engine successfully suggested the items in **Figure 29** as final recommendations to the user.

5.3 Mean Average Precision (MAP)

To evaluate the performance of our recommendation model we used the Mean average precision. This metric is used to measure how well the model predict relevant items and how accurately it can rank them. For each user a list of recommended items is generated and it calculates the precision at each point where a relevant item appears. It first finds the average of all the precision values and mean average precision is determined by finding the mean of the average values. The mean average precision was used because it focuses on precision, that is how many of the recommended items are relevant.

```
Mean Average Precision (MAP): 1.00
Actual Items Purchased by User: {'Jacket', 'Hoodie', 'Shirt', 'Sweater', 'Boots', 'Sneakers', 'Scarf'}
Relevant Items (Purchased by Others): {'Jacket', 'Shirt', 'Shoes', 'Dress', 'Hat', 'Jeans', 'Scarf', 'Shorts', 'Sandals', 'Pant
s', 'Blouse', 'Skirt', 'Handbag', 'Coat', 'Sweater', 'Boots', 'Gloves', 'Jewelry', 'T-shirt', 'Sunglasses', 'Backpack', 'Sneake
rs', 'Hoodie', 'Socks', 'Belt'}

['Belt', 'Jeans', 'Blouse', 'Jewelry', 'Shoes']
```

Figure 29 Mean Average Precision

Email	Item Purchased	Categor
lisagutierrez@johnson-herring.net	Scarf	Accessorie
lisagutierrez@johnson-herring.net	Scarf	Accessorie
lisagutierrez@johnson-herring.net	Sneakers	Footwear
lisagutierrez@johnson-herring.net	Shirt	Clothing
lisagutierrez@johnson-herring.net	Boots	Footwear
lisagutierrez@johnson-herring.net	Jacket	Outwear
lisagutierrez@johnson-herring.net	Sweater	Clothing
lisagutierrez@johnson-herring.net	Hoodie	Clothing

Figure 30

From **Figure 30** we checked the MAP for user “"lisagutierrez@johnson-herring.net”", the actual items purchased by the user was retrieved as shown in figure 30. In our purchased dataset.csv, we filtered the items purchased by the same user to compare as shown in **Figure 31**. They both matched, again the function retrieved the Relevant items. They relevant items are the items purchased by the neighbors. The function returned with Mean Average Precision

value of 1 as shown in **Figure30**. MAP value of 1 suggest a high-performance system where the recommended item for the user is relevant and the relevant items are perfectly ranked with the most relevant ones at the top of the list.

5.4 Conclusion

This chapter summarised the main project results obtained during the data understanding and modelling phases of the project. We successfully implemented a recommendation model and designed a recommendation engine to test the model. Also, we evaluated the performance of the model using Mean Average Precision (MAP). The model achieved a MAP of 1 which suggest a high-performance recommendation system.

Chapter 6

6.1 Conclusions and Future Works

This chapter discusses the conclusions drawn from this research and future works.

6.2 Conclusion

In this project recommendation model was developed using collaborative filtering and k-Nearest neighbor algorithm to suggest items to users based on their online engagement and purchase history. Below are the key considerations drawn from the research.

6.2.1 Model Optimisation and Cross-validation

The optimal value of k (nearest neighbors) was determined by testing a range of k values (from 1 to 20) using cross validation. The value of k that gave the highest accuracy was selected as the optimal k value for the final model. This approach provided an efficient model as it tested the model on different subset of data, leading to a more dynamic and reliable k value.

6.2.2 Feature Selection

The model utilises four key features, Time on app, Time on website, Length of membership and Yearly amount spent. These features correctly capture both the engagement metrics and spending behaviour, giving a holistic view of each customer

6.2.3 Personalised Recommendations

The recommendation engine built using k-NN algorithm was able to provide product recommendation to users. It first identifies the nearest neighbors using Euclidean distance, retrieves items purchased by the target user and the items purchased by the neighbors. It compares both and recommends most occurring items among the neighbor item list that are yet

to be purchased by the target user as final recommendation. This approach ensures that recommendation was relevant to the users' preferences and behaviour, enhancing the possibility of user engaging with the recommended product.

6.2.4 Cold Start Handling

Cold start is a significant challenge in recommendation systems in a situation where there is little or no prior data available for new users. This recommendation model handles the cold start problem by first checking if a user exists in the database before making a recommendation and if no it can switch other alternatives like recommending popular or trending items that are liked by the wider audience.

6.3 Future

To improve the performance of the recommendation model, future works should consider utilising additional features such user ratings into the recommendation system. For an ecommerce business user ratings promote customer engagement as it provides a platform for customers to share their experiences which can foster a sense of community and loyalty. User ratings can further enhance the accuracy of the model, it provides direct feedback on quality of items purchased. This can be used to offer personalised recommendations to other users.

Another key limitation of this recommendation model is that the data set did not capture user's gender. In ecommerce gender is a strong predictor of preferences in certain product categories like clothing, cosmetics and entertainment. However, it use must be balanced with careful attention to bias and privacy. Because our model did take into account the gender of users there are situations where female items were added to the recommendation list of a male user.

Lastly, future works can explore group recommendations, the algorithm can be expanded to recommend items for a group of users for example family shopping together or shared preferences among peers.

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Appendices

Appendices A Python Libraries

```
In [159]: import pandas as pd
          import numpy as np
          import seaborn as sns
          from  matplotlib import pyplot as plt
          from sklearn.preprocessing import  MinMaxScaler
          from sklearn.preprocessing import StandardScaler
          from sklearn.model_selection import train_test_split
          from sklearn.neighbors import NearestNeighbors
          from sklearn.neighbors import KNeighborsClassifier
          from sklearn.metrics.pairwise import cosine_similarity
          from sklearn.linear_model import LinearRegression
          from sklearn.metrics import accuracy_score
          from sklearn.model_selection import cross_val_score
          from sklearn.metrics import accuracy_score, precision_score
          from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
          import warnings
          import os
          import importlib
```

Appendices B Unified Dataset

	Email	Item Purchased	Category	Purchase Amount (USD)	Size	Color	Season	Payment Method
0	mstephenson@fernandez.com	Blouse	Clothing	53	L	Gray	Winter	Venmo
1	hduke@hotmail.com	Sweater	Clothing	64	L	Maroon	Winter	Cash
2	pallen@yahoo.com	Jeans	Clothing	73	S	Maroon	Spring	Credit Card
3	riverarebecca@gmail.com	Sandals	Footwear	90	M	Maroon	Spring	PayPal
4	mstephens@davidson-herman.com	Blouse	Clothing	49	M	Turquoise	Spring	PayPal